Classify Directly: A Dynamic Time SPA Classification Method Based on DTW

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***Abstract*—Side-channel analysis remains a critical threat to public-key cryptographic implementations. Simple Power Analysis (SPA) techniques can extract secret keys from a single power trace, often using clustering-based classification methods. However, traces captured in real-world environments often suffer from misalignment and variable trace lengths due to unstable clocks and random delays. As a result, clustering methods are required to use alignment methods that may alter the original information of the traces. To address this problem, this work proposes Dynamic Time Classification (DTC) as an alternative approach to classify cryptographic operations in SPA based on Dynamic Time Warping. Unlike clustering methods, DTC inherently compares power traces without requiring fixed-length segments, which greatly improved the adaptability to unequal traces and thus allows us to classify different operations relatively stably. Experimental results on public-key cryptographic algorithms and post-quantum algorithm implementations show that DTC are no less accurate than clustering methods and are more robust to timing variations. This work also systematically divides the features of different operations and explores the effects of different SPA methods on different types of feature. This work also conducts experiments with and without random delays for different categories, compares the experimental accuracy of different alignment methods, and discusses the feasibility of DTW as a preprocessing method.**

***Index Terms*—Side-Channel Analysis, Simple Power Analysis, Public-key Algorithms, Dynamic Time Warping, Kyber.**

I. Introduction

In the domain of integrated circuits, cryptographic algorithms are realized at the hardware level to ensure secure communication and data storage, with their security fundamentally dependent on the confidentiality and integrity of cryptographic key management within the chip. However, it has long been understood that securing the cryptographic algorithm itself is not sufficient—implementations must also resist attacks exploiting physical leakage, known as Side-Channel Analysis (SCA). SCA techniques extract secret keys by analyzing unintended physical emissions such as execution timing, power consumption, and electromagnetic radiation during cryptographic operations. The seminal work of Kocher in the 1990s introduced timing attacks and subsequently power analysis attacks, demonstrating that even mathematically secure algorithms can be compromised by observing these subtle physical characteristics [20].

Among various SCA techniques, Simple Power Analysis (SPA) [19] has attracted significant attention due to its remarkable simplicity and effectiveness [8]. Unlike Differential Power Analysis (DPA), which requires statistical analysis across numerous power traces, SPA allows an attacker to infer secret key bits from power traces by visual observation [12]. This makes SPA particularly concerning for public-key cryptographic algorithms—such as RSA [30] and Elliptic Curve Cryptography (ECC) [18, 24] —where long-term secret keys and conditional execution flows (e.g., square-and-multiply in RSA, point doubling and point addition in ECC) produce easily distinguishable power consumption patterns [22].

Early SPA techniques relied primarily on manual inspection of power traces, which was not only labor-intensive but also error-prone and impractical for sophisticated implementations. Meanwhile, in order to resist the previous attacks, the researchers proposed private key blinding making it impossible for attackers to use multiple traces to carry out attacks [4]. To automate and improve SPA, and in response to private key blinding, researchers increasingly adopted unsupervised clustering methods, leveraging algorithms such as K-means, DBSCAN, and hierarchical clustering, known as horizontal attacks [16]. These clustering-based methods enhance the efficiency of the analysis, improve the accuracy of the analysis, and are more conducive to the successful execution of the attack.

Despite their successes, clustering-based SPA methods encounter significant challenges. Specifically, the methods used by these techniques, including preprocessing methods using dimensionality reduction (e.g., PCA) [1] and different clustering algorithms, require explicit trace length normalization (e.g., padding). This means dealing with variable-length sub-traces (representing the segment of the trace that processes a single private key operation) due to, for example, unstable clock or random delays [37]. However, the choice of alignment methods is controversial and may artificially introduce bias or interference that can affect the accuracy of clustering. In addition, clustering methods are highly sensitive to measurement noise and misalignment [28], often limiting their effectiveness in real-world attack scenarios. Consequently, the practical application of clustering-based SPA remains limited by their dependency on precise alignment and uniform trace lengths.

Dynamic Time Warping (DTW), a classic algorithm initially developed for speech recognition [32], has been widely adopted in time-series alignment and classification tasks across various fields. In SCA, DTW has predominantly served as a preprocessing alignment method, aiding in aligning traces prior to statistical analyses. Despite its widespread success for trace alignment, DTW has not yet been extensively explored as a direct classifier for cryptographic operations within SCA.

In light of these observations, we propose Dynamic Time Classification (DTC), a novel approach utilizing DTW directly for classifying cryptographic operations in sub-traces.

Unlike existing clustering-based methods, our proposed DTC inherently accommodates variable-length sequences, naturally aligns misaligned sub-traces, and significantly reduces dependency on preprocessing steps. We also optimize the DTC method, which maintains high accuracy while significantly reducing computational overhead. Our systematic experiments across different algorithms and platforms, including RSA, ECC, and the post-quantum cryptographic algorithm Kyber, confirm DTC's superior accuracy, generalization against unstable clock and random delays, and wide applicability across different cryptographic implementations.

A. Related Works

Our work is related to a series of papers using the SPA approach. Kocher et al. [19] proposed the SPA method to make cracking public-key cryptographic algorithms simple and effective, and Courrege et al. [5] described in detail how the SPA method is used. To defend against SPA, Bajard et al. [2], Coron et al. [4], and Dupaquis et al. [7] propose countermeasures such as input messages blinding, exponentiation blinding, and modulo blinding. The core idea is to completely randomize the intermediate data during the algorithm. Thus, to combat blinding countermeasures, researchers have proposed horizontal attacks. Heyszl et al. [10] segmented traces by visual inspection or comparison of shifted trace parts, followed by an unsupervised K-means clustering applied to a single RSA or ECC exponentiation trace. This separates out the repeated operations and thus recover the exponent's bits. Guilherme Perin et al. [27, 26] successfully attacked the RSA cipher algorithm with blinding based on unsupervised learning by segmenting the traces equally and proposed a related framework. Jarvinen et al. [13] extended single-trace correlation attacks to scalar multiplication algorithms with precomputation by preprocessing through, for example, calculating averages, demonstrating that they can be attacked by clustering even when precomputation safeguards are used. Nascimento et al. [25] successfully attacked the ECC implementation of Montgomery Ladder on the ARM platform by segmenting the traces equally as well, proving the effectiveness of clustering attacks on embedded systems. Shi et al. [34] achieved the effective horizontal or single-trace attack on a practically protected RSA implementation by averaging the splitting of the traces on a modern microcontroller platform. Although the clustering algorithm works well, different papers do not explore the problems posed by time in detail, and tend to simply and roughly segment or not pay attention to it, and how this part is handled effects the accuracy of the experiment.

There are many ways to handle variable-length traces. One way is to use methods that are known a priori, so this part of the paper will not focus on them [29]. A more common way is visual inspection or comparison of shifted trace parts (overcutting or shortcutting) during trace segmentation [10, 36]. Another way to deal with variable-length traces is to align the traces after trace segmentation, Homma et al. [11, 35] proposed a matching method based on phase-only correlation (POC) to calculate the phase difference between two traces using cross-phase spectrum of the Discrete Fourier Transform of the traces. However, POC cannot combat countermeasures such as random delays or unstable clock when they cause time misalignment [23]. In this case, Diop et al. [6] use DTW for alignment. However, this approach has the potential to result in features being masked for SPA, which we will discuss in subsequent sections. These methods for dealing with unequal lengths are more or less flawed, so we aim to overcome the random delays, which is well addressed by DTC.

As mentioned previously, DTW-based "elastic alignment" has already been employed to synchronize power traces in SCA preprocessing by Van et al. [38]. Despite the fact that DTW have so many applications in SCA [15, 17, 14, 9, 21], few of them have been used in the classification of public-key cryptographic algorithm operations.

B. Our Contributions and Paper Organization

In this work, we propose DTC, a classification approach to effectively identify cryptographic operations from power traces based on DTW. The core contributions of our research can be summarized as follows:

- We propose DTC, a novel method that directly operates on raw, variable-length side-channel sub-traces without altering their original features. This method effectively mitigates the adverse effects of clock instability, random delays, and significant temporal misalignment commonly encountered in practical side-channel analysis.
- We model and divide the traces to be analyzed by SPA. By distinguishing the possible timing and features of the segmented sub-trace, we evaluate the advantages and disadvantages of different SPA methods on different categories of sub-traces and demonstrate the advantages and generalization of DTC.
- We validate the effectiveness and generalization of our proposed DTC methodology by conducting comprehensive experiments on various datasets, including different platforms power traces from RSA, ECC, and Kyber implementations, as well as by adding random delays to different datasets and discussing the effects of different alignment methods on the experimental results.

The rest of this paper is structured as follows: Section II provides background on SPA as well as comparison of DTW and Fast-DTW. Section III describes our proposed methodology for DTC. Section IV presents experimental results on RSA, ECC, and Kyber implementations and compares the accuracy of the DTC method with other methods, with and without random delay and with different alignment methods. Section V discusses the infeasibility of the DTW algorithm as a preprocessing approach to unequal trace lengths. Finally, Section VI concludes main findings and future research directions.

II. BACKGROUND

This section explores the basic techniques used for SCA, focusing on SPA as well as comparison of DTW and Fast-DTW.

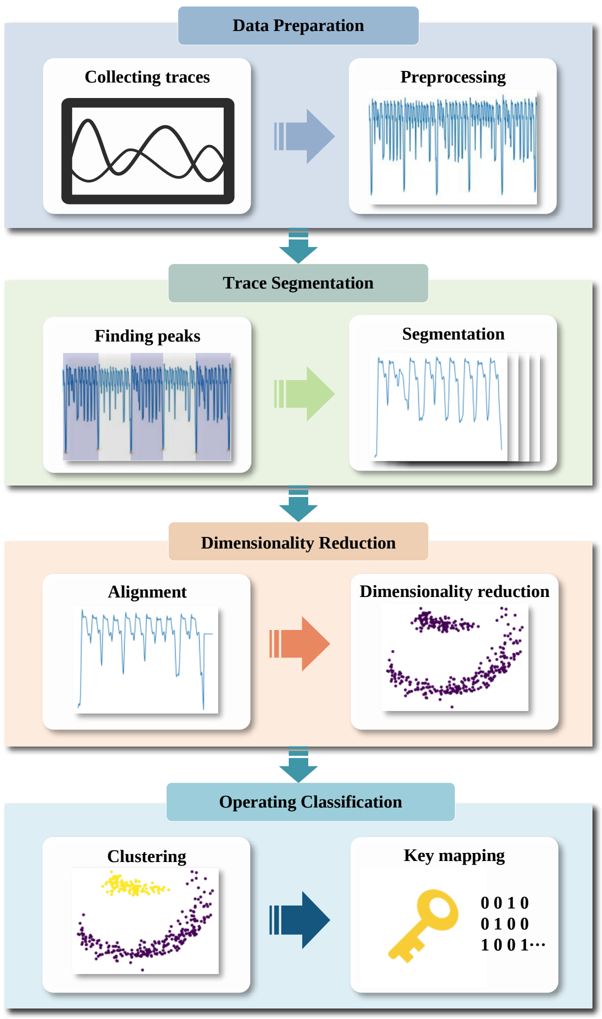


Fig. 1. Overall workflow of a public-key cryptographic algorithm operational identification method for traditional dimensionality reduction clustering.

A. SPA in Public-Key Cryptography

SPA leverages the fact that the execution of cryptographic operations typically involves distinguishable patterns of power consumption. In public-key cryptographic algorithms, such as RSA and ECC, secret key operations are often performed bit-by-bit, using conditional execution paths that depend explicitly on key bits. For instance, the RSA algorithm commonly employs the Square-and-Multiply method for modular exponentiation. In this method, each bit of the secret exponent triggers a distinct sequence of computational steps: if the bit is "1", the device performs a square operation followed by a multiplication; if the bit is "0", only a square operation is executed. Due to inherent differences in hardware implementation, these two operations exhibit unique and distinguishable power traces.

Nowadays, SPA methods mostly use methods such as dimensionality reduction clustering to circumvent problems such as manually interpreting traces that are error-prone and labor-intensive. As a result, more automated and effective approaches have emerged utilizing machine learning, statistical analysis, and unsupervised clustering techniques. As shown in Fig. 1, the method first prepares the data, i.e., collects the traces, preprocesses the traces, and then segments the traces using peaks or other methods. In the next step, these segmented traces are first aligned, thus making them of the same length before subjecting to a dimensionality reduction operation, and the results are clustered (e.g., K-means, DBSCAN, hierarchical clustering). Finally, the classification results are obtained based on the clustering results and mapped into key sequences. Our DTC method replaces the original trace alignment, dimensionality reduction, and clustering steps, making the overall process more concise.

B. Comparison of DTW and Fast-DTW

DTW is a classical and widely known algorithm originally developed to measure similarity between two temporal sequences that exhibit varying speeds, lengths, or timing offsets. The core advantage of DTW lies in its remarkable flexibility, allowing nonlinear temporal alignment by dynamically stretching or compressing the time axes of sequences. Due to this capability, DTW has been extensively employed in various domains such as speech recognition, gesture recognition, time-series classification, and general pattern matching problems, providing robust solutions to alignment challenges inherent in these applications.

Despite its strengths, the conventional DTW algorithm suffers from notable computational complexity, typically exhibiting quadratic complexity relative to the length of the sequences. This complexity quickly becomes prohibitive when applied to large-scale datasets or lengthy sequences, restricting its practical use in real-time or large-scale analytical scenarios.

To overcome the computational limitations of traditional DTW, the Fast Dynamic Time Warping (Fast-DTW) algorithm has been introduced. Fast-DTW utilizes a multi-resolution, hierarchical approach. It first compresses and aligns sequences at lower resolutions to quickly obtain an approximate alignment path. Subsequently, it progressively refines the alignment at higher resolutions. This hierarchical approximation significantly reduces computational requirements while maintaining a high level of alignment accuracy. As a result, Fast-DTW effectively extends the practical applicability of DTW, enabling its usage in scenarios involving larger datasets or situations demanding faster computation times. The principle of Fast-DTW is detailed in [33]. A comparison of DTW and Fast-DTW is shown in Fig. 2.

III. Methodology

In this section, we describe the overall flow of the DTC method as well as specific computational details. In order to verify the effectiveness of the DTC method, we also divide the traces of the operations in the cryptographic algorithm. The overall flow of our method is shown in Fig. 3.

A. DTC Distance Calculation

We define the DTC distance as the difference between the two sub-traces. We take the two sub-traces as A and B. The points on the two sub-traces are designated as A_i and B_j, the

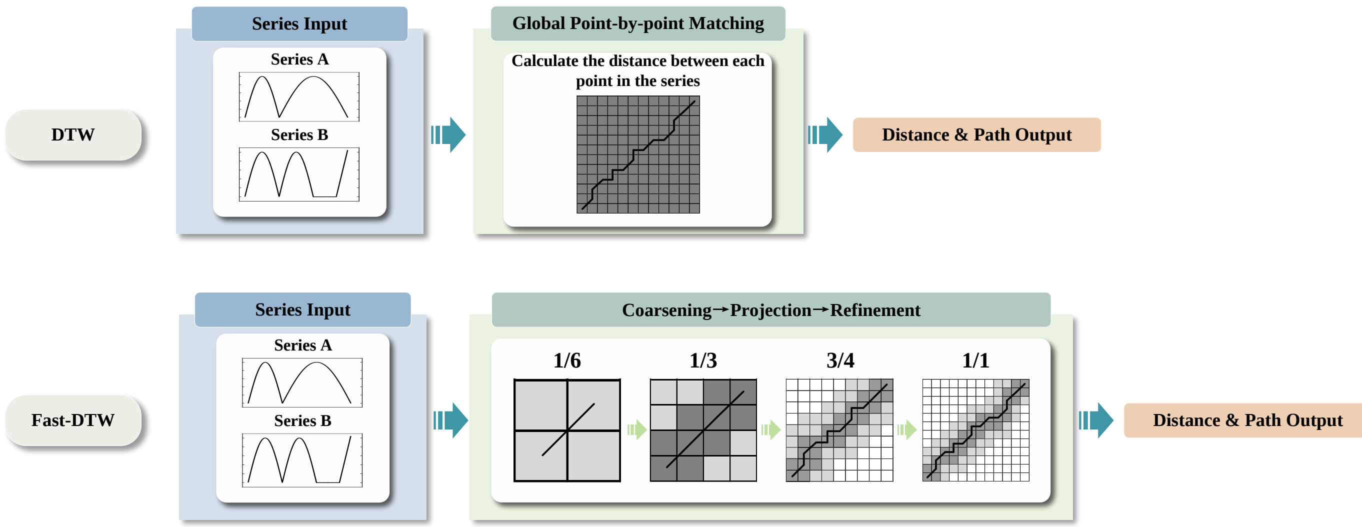


Fig. 2. Comparison of DTW and Fast-DTW algorithms. DTW directly calculates the finest resolution alignment and therefore has a higher computational complexity. Fast-DTW uses a multi-resolution strategy, starting with a coarse alignment at a lower resolution and iteratively improving the alignment, which greatly reduces the computational time.

lengths of A and B are subsequently defined as M and N, respectively. Subsequently, an $M \times N$ table can be constructed, wherein the horizontal and vertical axes represent the positions of the points of each sub-trace of A and B, and the values correspond to the distances between the corresponding points. Here the distance is used to reflect the positional relationship between the two sub-traces, which is usually calculated using the Euclidean distance, but also the Manhattan distance and other distances can be used. In order to save the time of square root operations, the distance is squared as:

$$d(i,j) = (A_i - B_j)^2 \tag{1}$$

where $d(i,j)$ is the distance between A_i and B_j. For instance, $d(1,1)$ denotes the distance value between the initial point of the sub-trace A and the initial point of the sub-trace B.

In order to calculate the DTC distance, we need to find a path that minimizes the cumulative distance from the bottom-left cell of the table to the top-right cell, as shown in Fig. 4. This path is denoted by W and starts at W_1 and ends at W_k. The cumulative distance is denoted by $D(i,j)$ and the cumulative distance of top-right cell is DTC distance. The exact computation will be given later.

It should first be noted that this path has not been chosen arbitrarily and must satisfy a number of constraints.

1) **Boundary:** The speed of articulation of any kind of speech may change, but the order of its parts cannot change, so the path chosen from $W_1 = (1,1)$ to $W_k = (M,N)$ must start from the lower left corner and end at the upper right corner.
2) **Continuity:** If $W_{i-1} = (a', b')$, then for the subsequent point in the path, W_i must satisfy the constraints $(a - a') \leq 1$ and $(b - b') \leq 1$. This implies that it is not feasible to align across a point, but rather to align with its immediate neighboring points. This ensures that every coordinate in A and B is represented in W.
3) **Monotonicity:** If $W_{i-1} = (a', b')$, then for the subsequent point in the path, W_i must satisfy the constraints $(a - a') \geq 0$ and $(b - b') \geq 0$. This ensures that the points above W are monotonic over time, preventing disruption between corresponding points in the time series.

By combining the continuity and monotonicity constraints, it can be seen that there are only three possible directions for the path of each point. To illustrate this, if the path has already passed through the point (i,j), then the next lattice point to pass through can only be one of the following three: $(i+1, j)$, $(i, j+1)$, or $(i+1, j+1)$.

Subsequently, the algorithm employs dynamic programming to perform the requisite calculations, utilizing the recursive formula:

$$D(i,j) = d(i,j) + min \begin{cases} D(i-1,j) \\ D(i,j-1) \\ D(i-1,j-1) \end{cases} \tag{2}$$

where $D(i,j)$ is the cumulative distance, and the cumulative distance at the top-right cell of the table is the DTC distance we need, calculate as:

$$DTC_{distance}(A,B) = D(M,N) = min\sqrt{\sum_{i=1}^{k} d(W_i)} \tag{3}$$

Because $M \neq N$ is allowed in the computation process, the DTC method is able to accommodate unequal sub-traces while keeping their original information.

B. Optimization

1) Fast Solution of DTC Distance: Motivated by Fast-DTW, we can reduce the computational complexity by adopting the idea of multi-resolution, coarsely downsampling the original time series layer by layer and using the same computational process to quickly find the approximate solution of

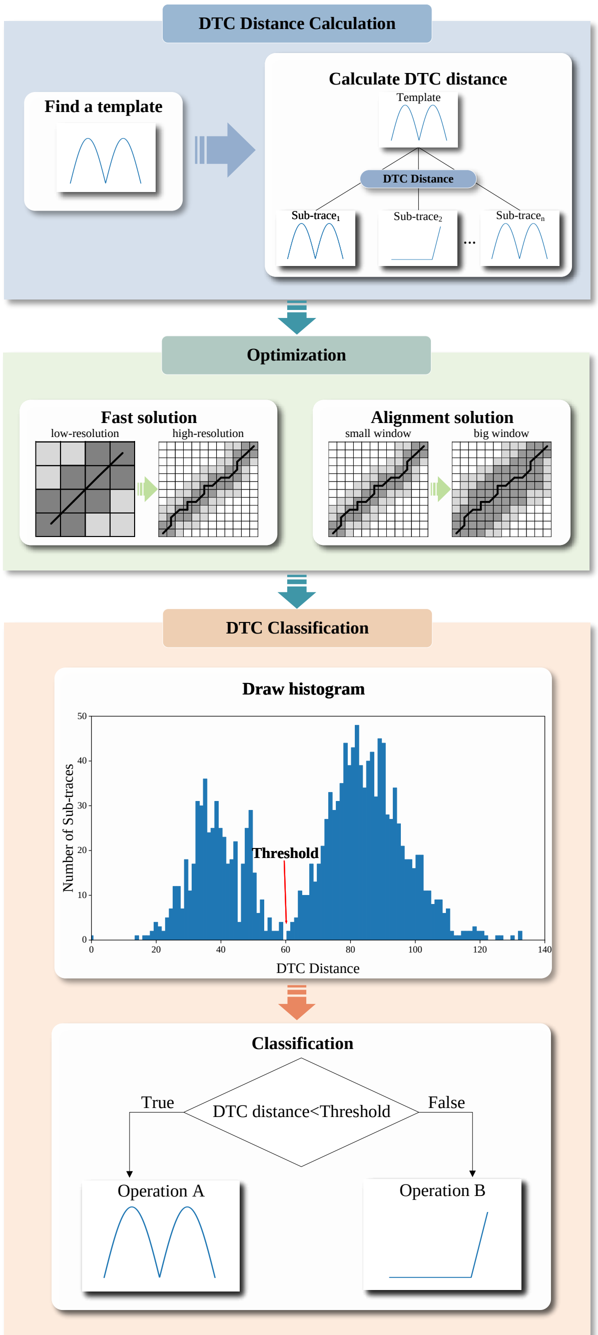


Fig. 3. The main workflow of public-key cryptographic algorithm operation classification method based on DTC method.

the optimal path in the low-resolution layer, and then gradually backtracking to the high-resolution layer to fine-tune the path search, which ensures high accuracy while greatly reduces the computational complexity. In practice, for two sequences of length M and N, the fast DTW algorithm, the fast solution reduces the computational complexity from $O(M \times N)$ to $O(N)$.

2) **Alignment Solution of DTC:** The DTC method seeks an optimal alignment that minimizes the total cumulative cost, focusing on the raw values of each point without considering their context or shape. Occasionally, noise can cause points from different operations that are distant in time to appear close in value. For example, a local peak might align with a trough because of their proximity in value, leading to a misleadingly small DTC distance. To mitigate this, we introduce the "window" strategy, which limits the warping range around the diagonal of the cumulative cost matrix. This adjustment, based on the noise level and temporal distortion of the trace, enhances the generalization and accuracy of the classification.

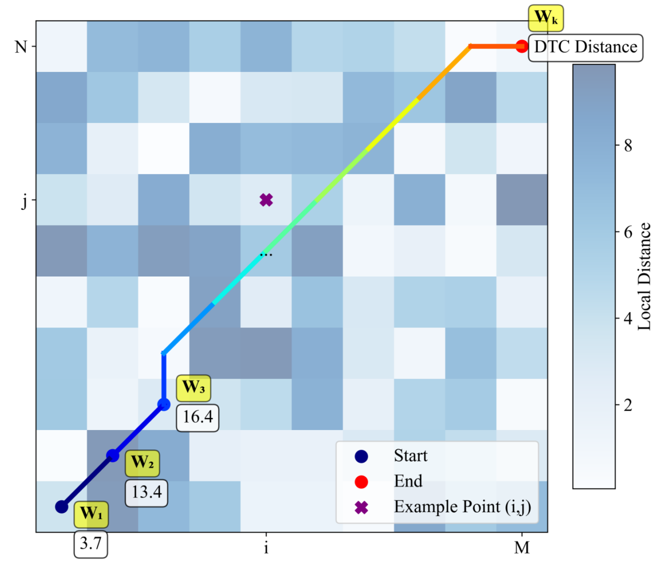


Fig. 4. Example of DTW algorithm for selecting warped paths. The purpose of the DTW algorithm is to select a path W such that the cumulative distance is minimized.

C. DTC Classification

We use the DTC method for classification in the following main steps:

- Select a sub-trace among the sub-traces after segmentation as a reference template.
- Calculate the DTC distance between each sub-trace and the template.
- Analyze the DTC distances calculated between all sub-traces and the template by plotting a histogram.

This histogram typically displays distinct peaks corresponding to different cryptographic operations, such as squaring and multiplication. By examining the distribution of distances, we can establish a threshold to differentiate operations. For instance, segments with distances below this threshold are highly similar to the template, while those above are less similar. Based on a set threshold, all sub-traces are classified, and different cryptographic operations are identified by classification and thus mapped to private keys.

D. Dividing the Categories of Different Operations

Different methods have their own strengths, however differences in equipment, acquisition environments, and encryption algorithms can produce different side information traces.

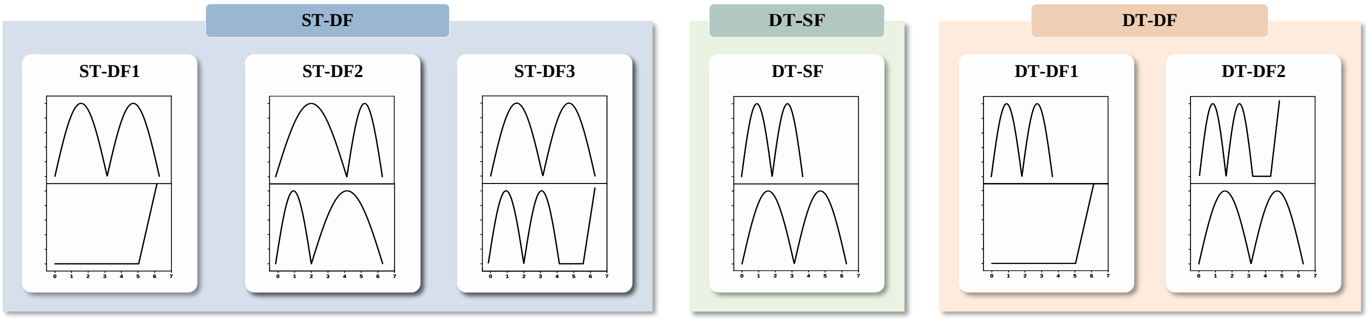


Fig. 5. Schematic diagram of dividing the categories of sub-traces corresponding to different operations after trace segmentation of public-key cryptography algorithm.

Therefore we made a systematic division. We divide sub-traces into three categories based on the timing and features of different operations after segmentation as shown in Fig. 5. Considering that there may be similar features within different operations but with different timing, we further subdivided these categories.

1) **Same Time, Different Features (ST-DF):** This category includes sub-traces that share identical timing but exhibit significantly different characteristic shapes, such as the amplitude of the energy and the distribution of the peak. In order to resist timing attacks, researchers have proposed constant time implementations, where any constant time implementation relies on fundamental constructs that implement certain basic operations in a constant time manner. Many implementations strive to equalize the timing of different operations in public-key cryptographic algorithms, which leads to varying features across implementations. To further evaluate the generalizability and effectiveness of DTC, we investigate this category in detail, subdividing it into three types:

- ST-DF1: Operations share the same timing, but features are completely uncorrelated.
- ST-DF2: Operations share the same timing, and one trace's features can be converted to another by warping.
- ST-DF3: Operations share the same timing, and one trace's features compress those of another.

2) **Different Time, Same Features (DT-SF):** This category includes traces that exhibit significant run-time differences between operations of the public-key cryptography algorithm, but overall features, such as shape and peak distribution, remain similar.

3) **Different Time, Different Features (DT-DF):** This final category includes traces where operations differ in timing and features, which are typically observed when the algorithm lacks protective measures. For a more detailed discussion of scenarios where the DTC method might falter, we further subdivide this category into two types:

- DT-DF1: Operations differ in timing and features, with no feature correlation.
- DT-DF2: Operations differ in timing and features, but some features of one trace overlap with those of another.

These categories target traces without random delays, and we will evaluate the advantages of different SPA methods on these categories and demonstrate the generalization of DTC algorithms.

IV. Experiments

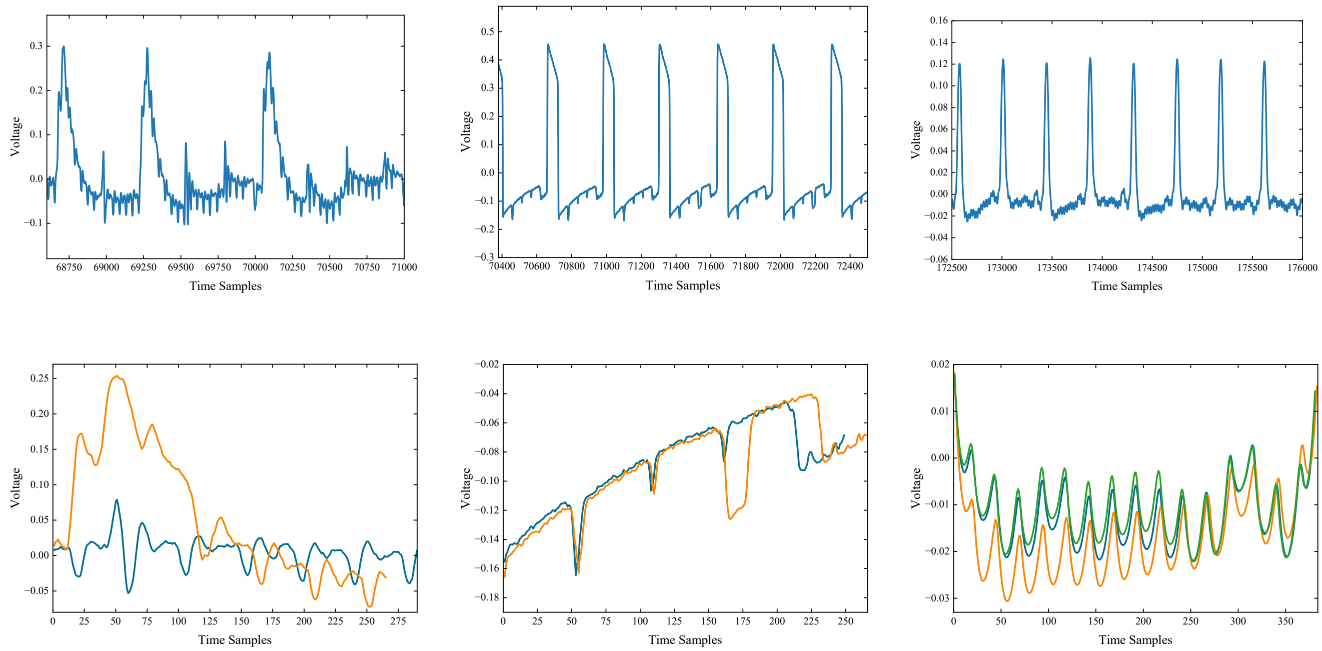
A. Experimental Setup

In this section, we evaluated the effectiveness of our proposed DTC method for classifying cryptographic operations with datasets of different algorithms and platforms. The information of the real datasets are shown in Table I. The "L_{key}" denotes the key length and "Operations" denotes the number of sub-traces after segmentation. Some of the traces are shown in Fig. 6 7 8 9. For RSA on AXIC X and ECC on AT89S52, the collected dataset will show multiple features, which we have also plotted in the figure. Post-quantum algorithms are a hot research topic nowadays, so we also analyze the Kyber algorithm [3]. We captured power traces from the Kyber algorithm implemented on the STM32F407 and mainly focus on the $poly\_frommsg()$ function used in decapsulation [39]. Since our captured real datasets could not cover all the division, we conducted simulation experiments for each of the five cases except ST-DF1, and the information of the simulated datasets are shown in Table II.

TABLE I
THE INFORMATION OF REAL DATASETS.

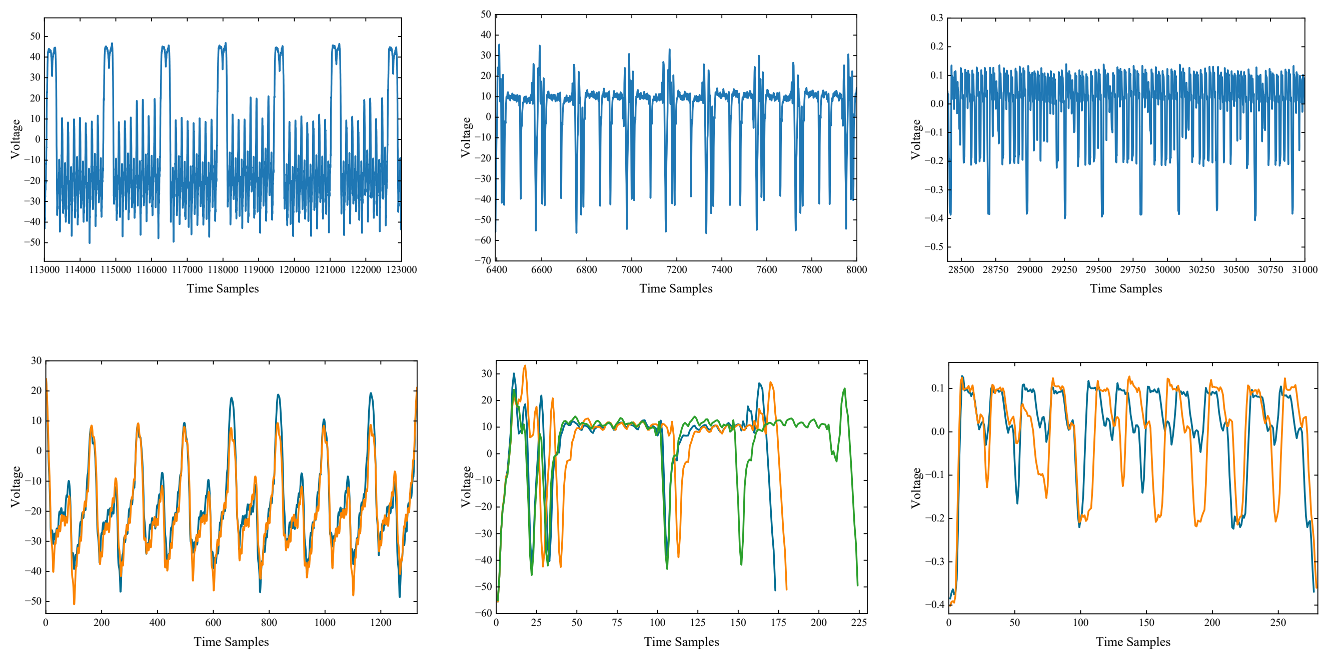
Algorithm	L_{key}	Operations	Device	Implementation
RSA	1024	1535	STM32F429	software
RSA	1024	1562	smart card	co-design
RSA	1024	1536	AXIC X	hardware
RSA	1024	1531	Sakura-G	hardware
ECC	128	192	AT89S52	software
ECC	256	365	smart card	co-design
Kyber	512	256	STM32F407	software

Our method is mainly compared with traditional dimensionality reduction and clustering methods. For dimensionality reduction we use PCA method, while for clustering we mainly use K-means and DBSCAN methods. Since the starting point of the K-means clustering method is randomly chosen and may have some uncertainty, our experiments count the average of the results of 1000 dimensionality reduction



(a) RSA on Sakura-G (b) RSA on smart card (c) RSA on AXIC X

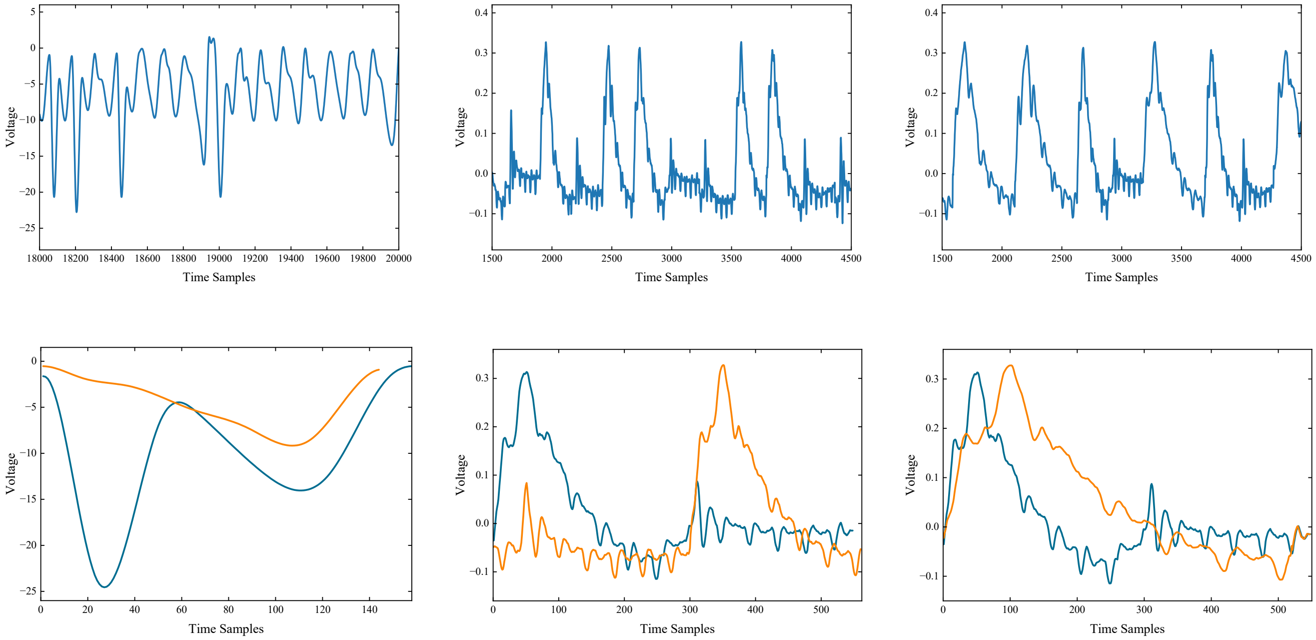
Fig. 6. Traces of datasets in different categories. The upper subfigures of the figure shows some of the continuous operation traces and the lower subfigure shows the traces of the different operations. (a) shows the RSA algorithm implemented on the STM32F429 platform, (b) shows the RSA algorithm implemented on the smart card platform, and (c) shows the RSA algorithm implemented on the AXIC X platform.



(a) RSA on Sakura-G (b) ECC on AT89S52 (c) ECC on smart card

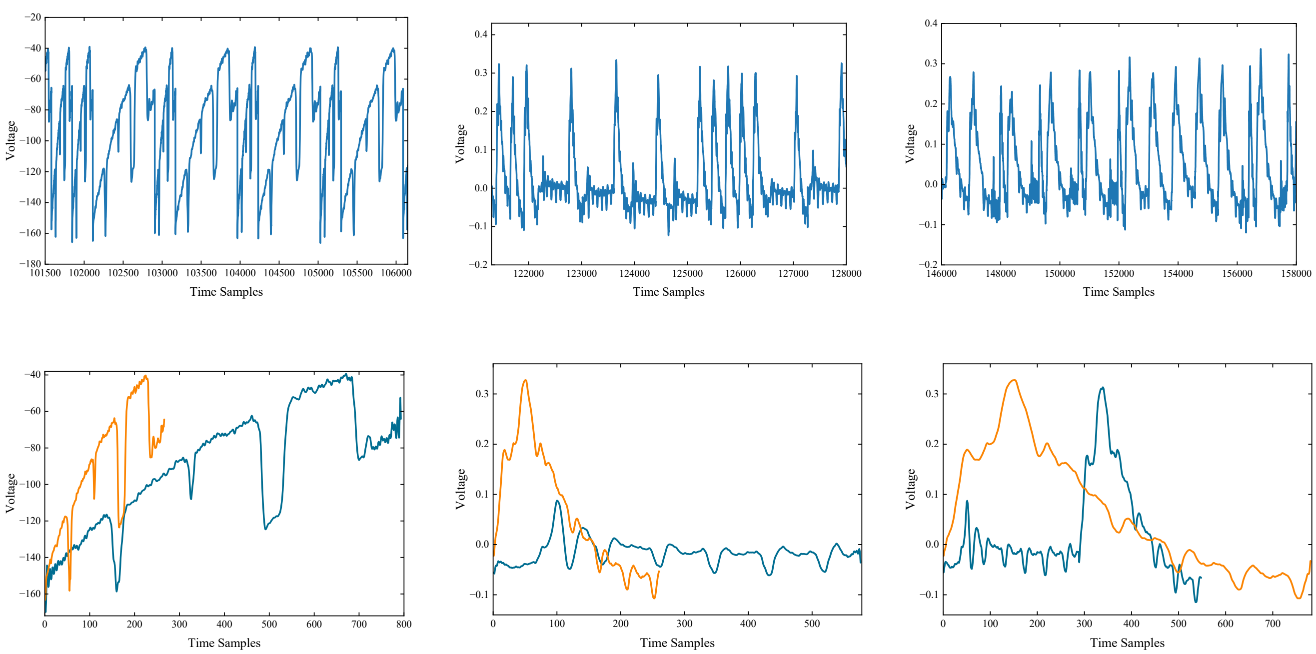
Fig. 7. Traces of datasets in different categories. The upper subfigures of the figure shows some of the continuous operation traces and the lower subfigure shows the traces of the different operations. (a) shows the RSA algorithm implemented on the Sakura-G platform, (b) shows the ECC algorithm implemented on the AT89S52 platform, and (c) shows the ECC algorithm implemented on the smart card platform.

clustering experiments. Since the selection of the radius of the DBSCAN algorithm also has a significant effect on the experimental results, we first found a better radius so that the feature points can be differentiated by a certain amount, and then we selected 1000 different radii in the vicinity of the better radius. The final accuracy of DBSCAN was the highest result among them. As the DTC uses distances between operations for comparison, our evaluations were also compared with results using the Manhattan and Euclidean distances directly. Specifically, we used the same template as



(a) Kyber on STM32F407 (b) Simulation dataset in ST-DF2 (c) Simulation dataset in ST-DF3

Fig. 8. Traces of datasets in different categories. The upper subfigures of the figure shows some of the continuous operation traces and the lower subfigure shows the traces of the different operations. (a) shows the Kyber algorithm implemented on the STM32F407 platform, (b) shows the simulated dataset on the ST-DF2 category, and (c) shows the simulated dataset on the ST-DF3 category.



(a) Simulation dataset in DT-SF (b) Simulation dataset in DT-DF1 (c) Simulation dataset in DT-DF2

Fig. 9. Traces of datasets in different categories. The upper subfigures of the figure shows some of the continuous operation traces and the lower subfigure shows the traces of the different operations. (a) showsthe simulated dataset on the DT-SF category, (b) shows the simulated dataset on the DT-DF1 category, and (c) shows the the simulated dataset on the DT-DF2 category.

in the DTC method and calculated the Manhattan or Euclidean distance between each sub-traces and the template, and finally plotted histograms for classification as well. For histograms where the threshold for classification is not well found, we choose the threshold based on the number of operations of the algorithm. The results of the DTC and the direct use of the Manhattan and Euclidean distances are always consistent when the experimental conditions are the same, so the results for these methods are for a single trial.

TABLE II
THE INFORMATION OF SIMULATED DATASETS IN OUR EXPERIMENTS

Division	L_{key}	Operations	Source
ST-DF2	1024	1536	RSA on STM32F429
ST-DF3	1024	1536	RSA on STM32F429
DT-SF	1024	1536	RSA on smart card
DT-DF1	1024	1536	RSA on STM32F429
DT-DF2	1024	1536	RSA on STM32F429

B. Experiments in DTC

We first conducted experiments on the datasets without random delays and the results are shown in Table III. For the clustering method, a zero-padding approach and PCA were used for preprocessing. The evaluation parameter is the accuracy of the recovery operation.

Experiments in ST-DF1. As the results shown in Table III, most of the datasets we have collected belong to this type due to most modern implementations taking measures to ensure that the running time is consistent across different operations. We observe that the accuracy of the different methods is high for RSA implemented on both the STM32F429 development board and the smart card platforms, indicating the correctness of the evaluation process. And through the experimental results of RSA implemented on ASIC X and Sakura-G, the correct rates of clustering methods are all lower than other methods. We found that the features of this dataset are mainly divided into three kinds due to the effect of noise, making the feature points after dimensionality reduction divided into three clusters, and the clustering methods have a high probability of misclassifying. However, the other methods do not face these challenges, making them more reliable in these cases. Therefore, in this case, we need to compare the experimental results of DTC with those of Manhattan distance and Euclidean distance.

Another noteworthy experimental result is the implementation of ECC on smart card, the results after PCA dimensionality reduction show that DBSCAN outperforms K-means, as well as Manhattan and Euclidean distances. Its performance was comparable to that of the DTC method. This highlights the fact that specific clustering algorithms can significantly outperform others when dealing with certain types of dimensionality reduction results. In this instance, the DTC's generalization allowed it to balance the advantages of clustering methods, making it equally effective.

The experimental results are promising: the classification accuracy of the DTC method is still higher than the rest of the algorithms even in the Kyber algorithm, which further validates its effectiveness in key recovery and classification tasks for various cryptographic algorithms.

Experiments in ST-DF2. During the evaluation of this type, although there are no traces of this type in the dataset we collected, we cannot guarantee that there are no such traces in reality and that the DTC algorithm will be effective on this type of dataset. Therefore, we conducted simulation experiments using trace data from the RSA algorithm implemented on the STM32F429 platform, as detailed in Table II. Specifically, we combined the modular squaring and modular multiplication operations of the RSA algorithm to create a new modular squaring operation; similarly, we combined the modular multiplication and modular squaring operations to create a new modular multiplication operation. In this way, we obtain a set of traces in which the modular squaring operation can be converted to a modular multiplication operation by warping.

According to the experimental results, the classification accuracy of the DTC method is on par with the rest of the method effects, and with the clustering method both reach 100% accuracy, proving that the DTC method is still effective in this classification, even though one trace's features can be converted to another by warping.

The experimental results show that the direct use of the Manhattan distance and the Euclidean distance does not reach 100%, and a comparison of their histograms with those of the DTC method is shown in Fig. 10. The classification threshold for the DTC algorithm is obvious, but the thresholds for the rest of the methods are not obvious. The accuracy of the DTC method is comparable to the effect of the other methods, and the classification accuracy with the clustering method reaches 100%, proving that the DTC method is still effective in this classification, even though the features of one trace can be converted to another by warping.

Experiments in ST-DF3. Similar to the ST-DF2 type evaluation process, we have no captured traces of this type. Therefore, we performed a similar simulation experiment, see Table II for details. Specifically, we resampled the modular squaring operation in the RSA algorithm at twice the frequency and used this result as a new modular squaring operation. We also combined the modular multiplication operation with the modular squaring operation to create a new modular multiplication operation. After our simulations we obtained a set of traces where the features of the modular multiplication operation contain the features of the modular squaring.

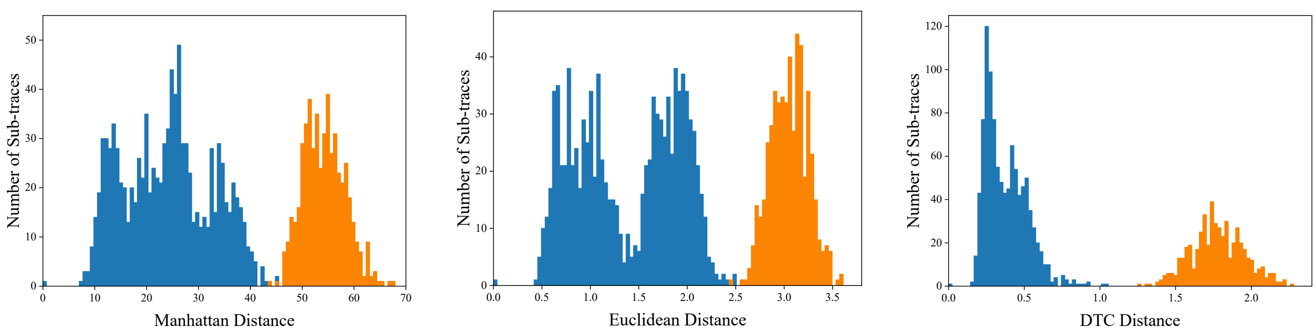
According to the experimental results, the classification accuracy of the DTC method is comparable to the effect of other methods, and the classification accuracy with the clustering method reaches 100%, proving that the DTC method is still effective in this kind of type even if there is compression of features between different operations.

Experiments in DT-SF. As mentioned earlier, the DTC method is an algorithm based on point-value similarity, which can be challenging when the point values of two different operations are very similar. In this case, the DTC distances between the same operation and different operations may be almost the same, which may lead to the failure of the algorithm. Therefore, it is necessary to verify whether the DTC method is still valid for this classification. Since we did not collect a real dataset for this classification, we conducted a simulation experiment using trace data from the RSA algorithm implemented on a smart card platform, details of which are given in Table II. Specifically, we used the modular squaring operation from the RSA algorithm as the new modular squaring operation, resampled this modular squaring operation at three times the frequency, and used these results as the new

TABLE III
EXPERIMENTAL ACCURACY RESULTS

Categories	Datasets	Manhattan	Euclidean	K-means	DBSCAN	DTC
ST-DF1	RSA on STM32F429	100%	100%	100%	100%	**100**%
	RSA on Smart Card	99.74%	99.74%	97.39%	99.68%	**99.74**%
	RSA on AXIC X	99.74%	99.74%	82.76%	66.64%	**99.80**%
	ECC on Smart Card	95.89%	94.79%	71.64%	100%	**100**%
	RSA on Sakura-G	85.37%	87.98%	75.73%	66.10%	**99.54**%
	Kyber on STM32F407	83.20%	82.81%	94.21%	93.75%	**100**%
ST-DF2	Sim. in ST-DF2[1]	99.80%	99.80%	100%	100%	**100**%
ST-DF3	Sim. in ST-DF3	100%	100%	100%	100%	**100**%
DT-SF	Sim. in DT-SF	100%	100%	100%	100%	**100**%
DT-DF1	Sim. in DT-DF1	100%	100%	100%	100%	**100**%
DT-DF2	ECC on AT89S52	100%	100%	74.43%	74.43%	**100**%
	Sim. in DT-DF2	100%	100%	100%	100%	**100**%

[1] Sim. denotes the Simulation dataset.



(a) Manhattan (b) Euclidean (c) DTC

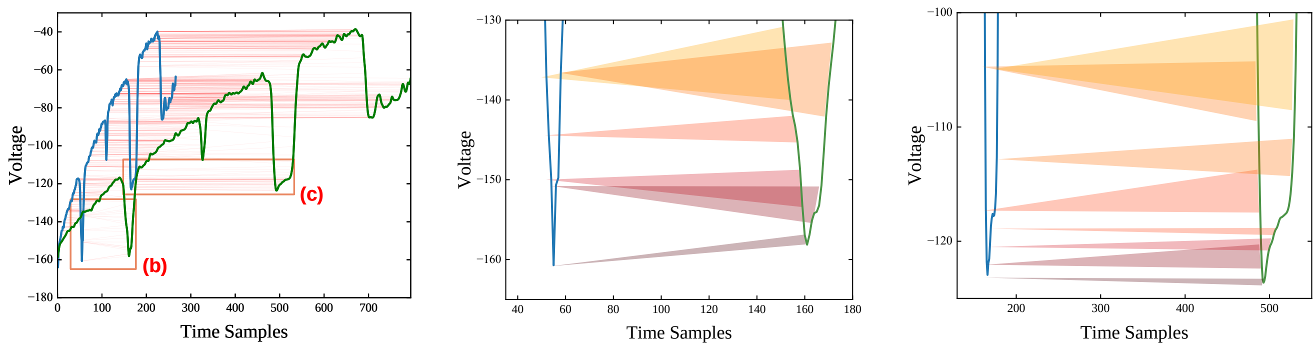
Fig. 10. Comparison of histograms using Manhattan distance, Euclidean distance and DTC method on ST-DF2 simulation dataset: (a) for using Manhattan distance, (b) for using Euclidean distance and (c) for using DTC method.

modular multiplication operation. This gives us a set of traces for which the characteristics of the modular squaring operation and the characteristics of the modular multiplication operation are the same but with different running times.

It is worth noting that the experimental accuracy of all methods was 100%. Based on these results, we can conclude that the DTC method did not fail in this type. We speculate that the success of the DTC method in this case may be attributed to the way it calculates the cumulative distance. Since the number of points varies from operation to operation, when we compute the DTC distance for operations of different lengths, we compute the distance between multiple points corresponding to a single point. Since the values between multiple points are similar but not identical to one point, the distance is not 0. This distance is likely to result in different DTC distances for the two operations. This is indeed the case, and we have investigated the specific process of calculating the DTC distance in this dataset, as shown in Fig. 11a, where the blue and green curves are the sub-trace and template to be calculated, and the red line is the distance between which two points the DTC distance has been specifically accumulated. We enlarged in on the two parts of the figure, as shown in Fig. 11b and Fig. 11c. For ease of viewing, we used different colors to indicate the points to be calculated. From the results in the figure, none of the slopes of the different connecting lines are 0, and therefore, as we hypothesized, their cumulative distances are not 0 either.

Experiments in DT-DF1. In evaluating this type, although there were no traces of this type in the dataset we collected, we suspected that this might be the case in situations involving unprotected devices with vulnerable code. Therefore, we conducted simulation experiments using trace data from the RSA algorithm implemented on the STM32F429 platform, see Table II for more details. Specifically, we used the modular multiplication operation in the RSA algorithm as a new modular squaring operation, and resampled the modular squaring operation at twice the frequency and used these results as a new modular multiplication operation. This gives us a set of traces with different timing and characteristics for the two operations.

The experimental results show that, similar to the results of the previous simulation experiments, the classification accuracy of the DTC method is comparable to the effectiveness of the other methods, and the classification accuracy using the clustering method reaches 100%. This also proves that the DTC method can be used in this type even if the time and



(a) Overall (b) Enlargement (c) Enlargement

Fig. 11. The process of calculating the DTC distance on the DT-SF simulation dataset: (a) shows the overall figure of the calculation, and (b) and (c) are enlarged figures of the corresponding parts of (a), where different colored connecting lines are used for easy differentiation.

features are different between the different operations.

Experiments in DT-DF2. In evaluating this type, the real dataset we collected, the ECC encryption algorithm implemented on the AT89S52 platform, fits this classification.

The experimental results from the AT89S52 dataset reveal that the success rate of both clustering methods is 74.43%. However, the DTC achieved a perfect 100% accuracy, as did the Manhattan and Euclidean distance methods. In this dataset, some of the traces were temporally shifted, causing their feature points to be divided into three clusters after dimensionality reduction, which led to the failure of the clustering algorithm. Similarly, Manhattan and Euclidean distances encountered issues with overlapping classes, though the distinction between similar classes was relatively small. In contrast, the DTC method performed robustly, unaffected by the temporal shifts. This experiment demonstrates that the DTW algorithm is resilient to traces that exhibit time offsets, providing more reliable classification results.

To further verify the effectiveness of the DTC under this classification, we also conducted simulation experiments. For this, we used the trace data from the RSA algorithm implemented on the STM32F429 platform, the details of which are given in Table II. Specifically, we resampled the modular multiplication operation in the RSA algorithm at three times the frequency, using these results as a new modular squaring operation. We then combined the modular squaring operation and modular multiplication operation to form a new modular multiplication operation. In this way, we obtained a set of traces with different operation runtimes but similar feature sections.

The experimental results show that all methods achieve 100% accuracy. These findings show that even in this type of situation, where the times of different operations are different but the features of one operation include the other, the DTC method still performs well and classifies effectively.

C. *Experiments with Random Delays*

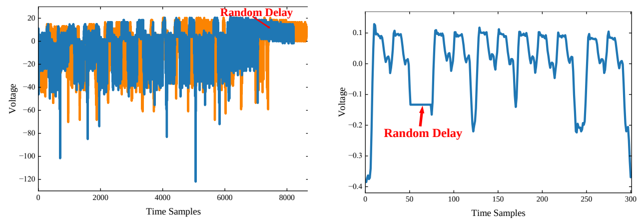
In the previous sections, we conducted experiments on both real and simulated datasets with different types to verify the generalization of the DTC method. These experiments were compared with dimensionality reduction and clustering approaches, as well as the direct use of Manhattan and Euclidean distances. The results showed that the DTC method performs excellently in many of the discussed types. However, all the datasets considered so far did not involve random delays. This countermeasure disrupts the proper alignment of data points, and as a result, the error from alignment traces to a uniform length becomes substantial, leading to erroneous results, regardless of whether dimensionality reduction clustering or distance-based methods are used. Roche et al. [31] manually removed the random delays when confronted with it, but it is clear that this method is more complicated.

One of the primary advantages of the DTW algorithm in SCA is its ability to mitigate the effect of random delays by aligning traces. This was one of the reasons we initially chose to experiment with the DTW-based algorithm, aiming to investigate its performance on datasets with random delays in public-key cryptography operations. Our goal was to minimize the impact of these random delays and assess the generalization of the DTC algorithm in this context.

To evaluate whether the DTC method continues to perform effectively in the presence of random delays, we implemented the ECC cryptographic algorithm on the Sakura-G development board, adding random delays after each operation, as shown in Fig. 12a. The experimental results shown in Table IV indicate that the accuracy of the DTC method is much higher than the rest of the methods and reaches 100%.

Since a single dataset is insufficient to conclusively demonstrate the advantage of the DTC method in the presence of random delays, we extended the experiments to cover one dataset from each of the six types discussed in previous chapters. In these experiments, we introduced random delays to the traces, simulating real-world conditions and further validating the DTW algorithm's ability to protect against the adverse effects of random delays. Fig. 12b shows an example of the addition of some of these random delays.

To ensure comprehensive protection, random delays were not limited to the end of each operation; instead, delays were introduced at various locations within the sub-traces. This was done to better reflect the random nature of delay insertion and to assess the algorithm's generalization. We added delays



(a) ECC on Sakura-G (b) ECC on smart card

Fig. 12. Comparison of random delays effects on ECC operations: real delays on Sakura-G and simulated delays on smart card.

of random magnitude and length at randomly chosen points within each sub-trace, ensuring that the delay maintained a similar energy profile to the previous data point at the insertion location, in accordance with the energy characteristics of the device.

The results, summarized in Table IV, demonstrate the varying impact of random delays on different methods. The application of random delays reduces the experimental accuracy of dimensionality reduction clustering. Notably, the direct use of Manhattan and Euclidean distances showed significant performance degradation due to the random delays. However, the DTC method remained remarkably resilient, with its accuracy staying close to 100% across most datasets, and achieving a perfect 100% accuracy for certain datasets. This further underscores the superiority of the DTC method in distinguishing between different operations across various classifications, even when random delays are introduced, in comparison to other methods.

D. Comparisons with Different Alignment Methods

In our previous evaluations, with the exception of the DTC method itself, which can handle data of varying lengths due to unstable clock or segmentation, all other methods require the filling of sub-traces immediately after segmentation in order to equalize their lengths before applying the various algorithms. As discussed earlier, the choice of alignment method for processing unequal sub-traces is a subject of debate. This is because no matter which alignment technique is selected, it invariably alters the original data to some extent to ensure uniform length, potentially compromising the effectiveness of the classification algorithms. However, the DTC method, by accepting data of varying lengths, bypasses the need for alignment and preserves the integrity of the data, thereby often achieving more accurate classification results. This inherent advantage of the DTC method prompts us to explore the impact of different alignment methods on experimental outcomes in this subsection.

To ensure a comprehensive evaluation, we selected datasets from each of the six types discussed previously, both with and without random delays. Furthermore, we focused on those experiments where the DTC method demonstrated significantly higher accuracy compared to the combination of dimensionality reduction and clustering, to determine whether the alignment method or the classification algorithm itself influences these results.

The common alignment methods we considered include zero padding, overcutting, undercutting, and interpolation. Historically, in all our previous experiments, we employed zero padding, adding zeros at the end of the sub-traces to standardize their lengths. This method is straightforward and widely used, but not always the most effective. The experimental results of the zero padding are shown in Table III and Table IV, respectively. Moving forward, we will conduct experiments using overcutting, undercutting, and interpolation as alignment strategies to further investigate their effects on the classification accuracy of the datasets.

Comparison with overcutting. The overcutting method involves extending each sub-trace to match the maximum length observed in the dataset. For real datasets, we maintain the start of each sub-trace constant and extend backwards until reaching the maximum length. For simulated datasets, we append subsequent operational sub-traces at the end of each sub-trace. This method risks introducing irrelevant segments and additional noise, potentially degrading the classification accuracy.

The results are shown in Table V, our experimental results show a general decline in accuracy using the overcutting method compared to zero-padding, with only minor improvements observed in specific datasets. Our DTC method is more accurate than methods that use this alignment method. This outcome occurs because overcutting often includes intervals of different operations that contain identical spikes and noise, thereby adding extraneous data and reducing overall accuracy.

Comparison with undercutting. Undercutting involves trimming longer sub-traces to match the shortest one within the dataset. This approach ensures the original signal's integrity by avoiding the addition of artificial data, though it risks losing significant data points.

The results are shown in Table VI, the accuracy of the undercut method varied across different datasets. In cases where crucial distinguishing features are located at the end of subtraces, undercutting may lead to a loss of important information, decreasing accuracy. Conversely, if the trimmed sections primarily contain noise, removing them could improve accuracy. Although some of the datasets have improved in accuracy compared to the alignment approach using complementary zeros, the accuracy of our DTC method remains equal to or higher than that method.

Comparison with interpolation. Interpolation adjusts subtrace lengths by creating smooth transitions between data points, aiming for uniformity across all traces. This method generally preserves more original data features than other alignment techniques. In our approach, we resample sub-traces until their lengths align with the longest sub-trace. However, this method can obscure the temporal differences between operations and might introduce temporal shifts in features, potentially leading to reduced accuracy.

The results are shown in Table VII, our experiments indicate that interpolation has distinct advantages, particularly noted in the ECC cryptographic algorithm on the AT89S52 platform, where accuracy improved significantly. Zero-padding led to data being clustered into three primary groups, resulting in lower clustering accuracy. However, interpolation resolved

TABLE IV
EXPERIMENTAL ACCURACY RESULTS WITH RANDOM DELAYS

Categories	Datasets	Manhattan	Euclidean	K-means	DBSCAN	DTC
ST-DF1	ECC on Smartcard	64.56%	62.36%	65.15%	67.31%	**99.18**%
ST-DF2	Sim. in ST-DF2	69.86%	75.56%	99.02%	96.09%	**99.93**%
ST-DF3	Sim. in ST-DF3	66.99%	74.67%	75.72%	80.01%	**96.81**%
DT-SF	Sim. in DT-SF	88.02%	89.97%	95.77%	98.31%	**99.61**%
DT-DF1	ECC on Sakura-G	70.83%	71.35%	73.18%	74.74%	**100**%
	Sim. in DT-DF1	78.45%	82.62%	87.98%	94.01%	**99.93**%
DT-DF2	Sim. in DT-DF2	92.38%	96.42%	98.96%	97.33%	**100**%

TABLE V
EXPERIMENTAL ACCURACY RESULTS USING THE OVERCUTTING METHOD

Categories	Datasets	Random Delays	Manhattan	Euclidean	K-means	DBSCAN	DTC
ST-DF1	RSA on Sakura-G	No	87.27%	92.36%	73.51%	66.78%	**99.54**%
	RSA on ASIC-X	No	94.13%	70.12%	61.34%	94.32%	**99.80**%
	ECC on Smartcard	Yes	57.46%	57.46%	59.80%	67.68%	**99.18**%
ST-DF2	Sim. in ST-DF2	No	73.53%	73.01%	77.57%	77.57%	**100**%
		Yes	56.27%	58.22%	57.46%	66.71%	**99.93**%
ST-DF3	Sim. in ST-DF3	No	77.58%	77.44%	77.57%	77.57%	**100**%
		Yes	62.92%	63.05%	53.17%	66.64%	**96.81**%
DT-SF	Sim. in DT-SF	No	58.41%	58.69%	55.07%	35.85%	**100**%
		Yes	74.79%	72.83%	68.13%	66.75%	**99.61**%
DT-DF1	Sim. in DT-DF1	No	56.29%	57.60%	55.78%	66.73%	**100**%
		Yes	57.01%	56.75%	60.72%	66.67%	**99.93**%
	ECC on Sakura-G	Yes	56.02%	56.02%	55.50%	55.76%	**100**%
DT-DF2	ECC on AT89S52	No	82.89%	82.89%	74.77%	56.68%	**100**%
	Sim. in DT-DF2	No	77.51%	77.51%	77.57%	77.57%	**100**%
		Yes	76.53%	78.10%	82.74%	81.62%	**100**%

TABLE VI
EXPERIMENTAL ACCURACY RESULTS USING THE UNDERCUTTING METHOD

Categories	Datasets	Random Delays	Manhattan	Euclidean	K-means	DBSCAN	DTC
ST-DF1	RSA on Sakura-G	No	82.44%	90.27%	71.94%	66.51%	**99.54**%
	RSA on ASIC-X	No	99.61%	99.74%	84.75%	66.67%	**99.80**%
	ECC on Smartcard	Yes	73.42%	71.23%	61.90%	67.40%	**99.18**%
ST-DF2	Sim. in ST-DF2	No	99.80%	99.80%	100%	100%	**100**%
		Yes	93.55%	92.64%	98.96%	95.38%	**99.93**%
ST-DF3	Sim. in ST-DF3	No	99.93%	99.93%	100%	100%	**100**%
		Yes	72.85%	75.72%	78.13%	79.02%	**96.81**%
DT-SF	Sim. in DT-SF	No	100%	100%	100%	100%	**100**%
		Yes	88.28%	84.38%	77.76%	66.67%	**99.61**%
DT-DF1	Sim. in DT-DF1	No	99.93%	99.93%	100%	100%	**100**%
		Yes	95.64%	96.29%	98.11%	99.09%	**99.93**%
	ECC on Sakura-G	Yes	70.31%	71.35%	68.91%	89.84%	**100**%
DT-DF2	ECC on AT89S52	No	79.68%	82.89%	74.86%	41.71%	**100**%
	Sim. in DT-DF2	No	100%	100%	100%	100%	**100**%
		Yes	97.72%	98.24%	99.02%	98.31%	**100**%

TABLE VII
EXPERIMENTAL ACCURACY RESULTS USING THE INTERPOLATION METHOD

Categories	Datasets	Random Delays	Manhattan	Euclidean	K-means	DBSCAN	DTC
ST-DF1	RSA on Sakura-G	No	85.44%	90.67%	75.66%	69.39%	**99.54**%
	RSA on ASIC-X	No	99.54%	99.54%	79.69%	66.64%	**99.80**%
	ECC on Smartcard	Yes	67.40%	69.04%	57.81%	66.30%	**99.18**%
ST-DF2	Sim. in ST-DF2	No	99.93%	99.93%	100%	100%	**100**%
		Yes	90.56%	91.86%	99.54%	97.53%	**99.93**%
ST-DF3	Sim. in ST-DF3	No	99.93%	99.93%	100%	100%	**100**%
		Yes	90.04%	89.13%	89.28%	78.13%	**96.81**%
DT-SF	Sim. in DT-SF	No	55.73%	55.99%	50.94%	66.41%	**100**%
		Yes	76.43%	73.05%	54.40%	64.32%	**99.61**%
DT-DF1	Sim. in DT-DF1	No	99.93%	99.93%	100%	100%	**100**%
		Yes	96.29%	94.99%	97.11%	99.09%	**99.93**%
	ECC on Sakura-G	Yes	65.63%	67.19%	98.51%	66.41%	**100**%
DT-DF2	ECC on AT89S52	No	100%	100%	97.84%	100%	**100**%
	Sim. in DT-DF2	No	100%	100%	100%	100%	**100**%
		Yes	99.80%	99.80%	99.80%	99.41%	**100**%

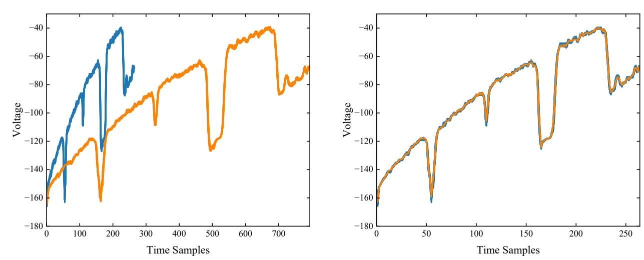
this issue. Interestingly, in the DT-SF classification, interpolation's removal of temporal differences significantly lowered accuracy, underscoring a drawback of this method in certain scenarios.

Upon reviewing various alignment methods, it's clear that no single approach consistently outperforms others across all datasets and classifications. The optimal alignment method depends on the specific dataset and the nature of the data. While different alignment methods can yield similar accuracies within the same dataset, choosing the best method requires extensive experimentation. However, the DTC method, which accepts data of unequal lengths, consistently shows robust performance without the need for alignment, further validating its effectiveness. Even when different alignment methods enhance the accuracy of other techniques, the DTC method maintains, or even surpasses, its performance level, emphasizing its utility and reliability.

V. DISCUSSION

In the previous sections, we evaluated the effectiveness of our proposed DTC method in categorizing cryptographic operations on datasets with different algorithms and platforms, respectively, and we verified that the accuracy of the DTC method is good with the addition of random delays as well as with the use of different alignment methods. One might ask why we do not use DTW as a preprocessing approach to align sub-traces before applying the dimensionality reduction and clustering methods. For this reason, we discuss this question experimentally in this section.

The DTW algorithm is generally used to deal with unequal-length traces, so a dataset with and without random delays is selected for experimentation from each of the different time categories. We mainly compare the results of the method using the DTW algorithm as preprocessing for dimensionality reduction clustering with the DTC method, and the experimental results are shown in Table VIII.



(a) Before DTW (b) After DTW

Fig. 13. Comparison of traces before and after DTW on the DT-SF dataset.

From the experimental results, it is found that the accuracy of the experiments on most of the data sets is not high, and the explanation of this result comes from the working mechanism of the DTW algorithm. According to its principle, the DTW algorithm selects a sub-trace from the data sets as a template. The DTW algorithm is then used to align subsequent sub-traces with this template. This process can cause sub-traces from different operations to converge on the template, making them virtually indistinguishable. Unless the difference in features between the different sub-traces is very obvious, as in the case of the Sim. in DT-DF1. Like in the DT-SF classification, the only difference between the different operations is their time, while their features are almost identical. When the DTW algorithm is applied in this case, it adjusts the lengths of the sub-traces so that they match, making the different operations look similar as shown in Fig. 13. This effect masks the different features needed to distinguish the different operations, making them indistinguishable after alignment. This explains why, although DTW is effective in dealing with random delays, using DTW for alignment before clustering can degrade the ability to

TABLE VIII
COMPARISON OF EXPERIMENTAL RESULTS USING DTW AS A PREPROCESSING METHOD WITH DTC METHOD

Categories	Datasets	Random Delays	K-means	DBSCAN	DTW+K-means	DTW+DBSCAN	DTC
DT-SF	Sim. in DT-SF	No	100%	100%	60.77%	66.67%	**100**%
		Yes	95.77%	98.31%	60.63%	66.67%	**99.61**%
DT-DF1	Sim. in DT-DF1	No	100%	100%	99.41%	100%	**100**%
	ECC on Sakura-G	Yes	73.18%	74.74%	66.67%	52.60%	**100**%
DT-DF2	ECC on AT89S52	No	74.43%	74.43%	58.06%	66.79%	**100**%
	Sim. in DT-DF2	Yes	98.96%	97.33%	52.31%	66.67%	**100**%

distinguish between different cryptographic operations.

VI. CONCLUSION

In this paper, our main contribution is the proposal of the DTC method recognize the operation of public-key cryptosystems. This study shows that the DTC method adapts to the stretching and compression of unequally long time series by maintaining the integrity of the data through warping, and thus performs well in calculating the matching degree of time series of unequal lengths. And it verifies that DTW can be used not only as a preprocessing tool but also as an effective classifier for SCA. Furthermore, the applicability of the DTC approach is not limited to simple analysis; it is also robust to measures such as random delays, which are designed to protect cryptographic operations from SCA.

In addition, we systematically classify the feature differences of operations within public-key cryptography algorithms and explore the effect of different SPA methods on different featured data, and in these experiments, DTC exhibits generalization. We have used different trace alignment methods for the dimensionality reduction clustering experiments and compared the results with the DTC method, which further validates the effectiveness of the DTC algorithm. We also used the DTW algorithm as a preprocessing method and experimented again with unequal length traces including with and without random delays. The experiments verified that using the DTW algorithm as an alignment method in preprocessing does not work well in cryptographic algorithm operation classification.

In the future, there is a need to improve DTC methods to increase speed and reduce computational complexity. In addition, the problem of classification in the presence of large amounts of noise needs to be addressed. Addressing these issues will not only refine the DTC method, but also improve its adaptability and accuracy in more challenging situations.

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